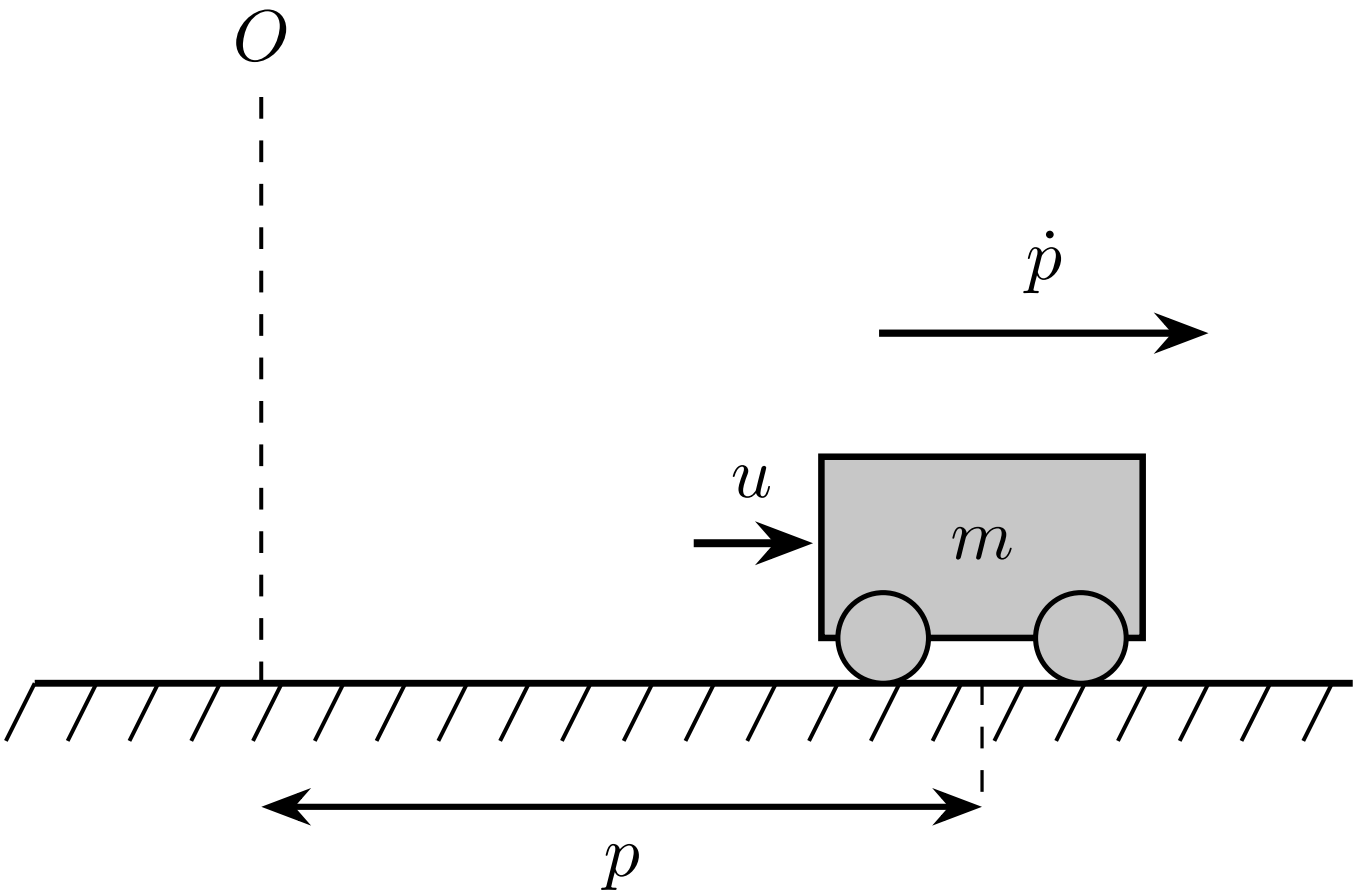

$$x = \begin{bmatrix} p \\ \dot{p} \end{bmatrix} \in \mathbb{R}^2$$

$$\ddot{p} = u$$